Discrete Mathematics - Practical 1

Set Theory: worked solutions

with the reasoning spelled out, step by step

Covers the assigned subset of Practical 1: problems 2, 7, 11, 16, 20, 54 (parts 3 and 11), 58, 60.

Warm-up

This sheet is mostly about *proving* set statements, not computing numbers. Three kinds of task appear. (1) **Membership puzzles** (problems 2, 7, 11): keep a clear head about the difference between an element x and the one-element set $\{x\}$ that contains it; $\in$ and $\subseteq$ are different relations. (2) **Set identities** (16, 20, 54, 60): the universal tool is *element-chasing* - to prove $X = Y$, show that an arbitrary element lies in X if and only if it lies in Y, by rewriting the defining condition step by step. (3) A small **construction / structural** argument (58).

Notation used here: $\overline{A}$ is the complement of A (everything in the universe U that is not in A); $A \setminus B$ is set difference (in the source sheet it is written $A - B$); $A \triangle B$ is the *symmetric difference* (written $\div$ in the source), defined by $A \triangle B = (A \setminus B) \cup (B \setminus A)$; and $\subset$ means *strict* inclusion ($A \subseteq B$ and $A \neq B$), while $\subseteq$ allows equality.

The single most useful habit on this sheet: when stuck, drop down to the level of one element. Almost every line below is just "what does it mean for a fixed x to belong here?"

Problem 2 - Which membership statements are true?

Problem 2

Decide which of the following are true.

2.1 $\varnothing \in \{\varnothing, \{\varnothing\}\}$

2.2 $\{\varnothing\} \in \{\{\varnothing\}\}$

2.3 $\varnothing \in \{\{\varnothing\}\}$

The idea

$x \in S$ is true exactly when x is *literally one of the items listed inside the braces of* S - nothing more. So read each set as a list and ask: is the thing on the left one of the entries on the right? The only trap is that the entries can themselves be sets, and you must not confuse $\varnothing$ (the empty set, 0 elements) with $\{\varnothing\}$ (a box containing the empty set, 1 element). They are different objects, so $\varnothing \neq \{\varnothing\}$. Treat $\varnothing$ and $\{\varnothing\}$ as two distinct "names" and just scan the list.

Step 1 - List the entries of each right-hand set.

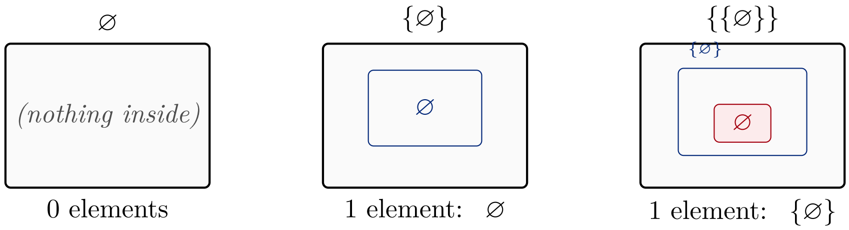
- $\{\varnothing, \{\varnothing\}\}$ has *two* entries: $\varnothing$ and $\{\varnothing\}$.
- $\{\{\varnothing\}\}$ has *one* entry: $\{\varnothing\}$.

Step 2 - Check each claim against the list.

- **2.1** $\varnothing \in \{\varnothing, \{\varnothing\}\}$: is $\varnothing$ one of the two entries? Yes, it is the first one. **True.**
- **2.2** $\{\varnothing\} \in \{\{\varnothing\}\}$: is $\{\varnothing\}$ the single entry? Yes, the entry is exactly $\{\varnothing\}$. **True.**
- **2.3** $\varnothing \in \{\{\varnothing\}\}$: is $\varnothing$ the single entry? The only entry is $\{\varnothing\}$, and $\varnothing \neq \{\varnothing\}$ (zero elements versus one). So no. **False.**

Always check. A quick sanity test that 2.3 is genuinely false: $\{\{\varnothing\}\}$ has exactly one element. If both $\varnothing$ and $\{\varnothing\}$ belonged to it, it would have two distinct elements. It does not, so at most one of them is in it - and 2.2 already pinned that one down as $\{\varnothing\}$. ✓

The three objects as nested boxes (a box is a set; what sits loose inside it are its elements). Reading them side by side shows why they are all different:



Answer: 2.1 True, 2.2 True, 2.3 False.

★ Method to remember

Transferable technique: $x \in S$ asks only "is x an item in the top-level list of S?". Do not look inside the items. And always keep x separate from $\{x\}$: wrapping something in braces makes a new, different object with one element.

Problem 7 - $\in$ is not transitive

Problem 7

Give sets A, B, C with $A \in B$, $B \in C$, but $A \notin C$.

The idea

For relations like $\leq$ or $\subseteq$, chaining works: $a \leq b \leq c$ forces $a \leq c$. The point of this exercise is that *membership* $\in$ *does not chain.* $A \in B$ says A sits one level down inside B; $B \in C$ says B sits one level down inside C, so A is now *two* levels down inside C - which is a completely different statement from "A is an element of C". The cleanest way to feel this is to build a tower where each set is wrapped once more in braces. The classic tower is $\varnothing$, $\{\varnothing\}$, $\{\{\varnothing\}\}$.

Step 1 - Choose the tower. Let

$$A = \varnothing, \qquad B = \{\varnothing\}, \qquad C = \{\{\varnothing\}\}.$$

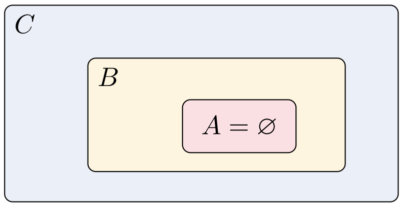
Step 2 - Verify the two memberships.

- $A \in B$: is $\varnothing$ an entry of $\{\varnothing\}$? Yes. ✓
- $B \in C$: is $\{\varnothing\}$ an entry of $\{\{\varnothing\}\}$? Yes, it is the only entry. ✓

Step 3 - Verify the non-membership. $C = \{\{\varnothing\}\}$ has a single entry, $\{\varnothing\}$. Is $A = \varnothing$ that entry? No, $\varnothing \neq \{\varnothing\}$. So $A \notin C$. ✓

Answer: $A = \varnothing$, $B = \{\varnothing\}$, $C = \{\{\varnothing\}\}$. This witnesses that $\in$ is not transitive.

Picture of the nesting (each box is one level; A is buried two levels deep inside C, so it is not a direct entry of C):



$A \in B$, $B \in C$

but $A \notin C$

Problem 11 - A mixed inclusion / membership chain

Problem 11

Give sets A, B, C, D, E with $A \subset B$, $B \in C$, $C \subset D$, $D \subset E$, where every $\subset$ is strict.

The idea

The puzzle mixes two different relations in one chain, and the only awkward link is the middle one, $B \in C$: here B must appear as a whole *element* of C, not as a subset. So engineer C so that one of its listed entries is exactly the set B. Everything else is routine: for a strict inclusion $X \subset Y$, just take Y to be X with one extra element thrown in. Build $A \subset B$ first, then make C a set that *contains B as an item*, then keep adding one element to climb $C \subset D \subset E$.

Step 1 - The strict pair $A \subset B$. Take $A = \{1\}$ and $B = \{1, 2\}$. Then $A \subseteq B$ and $A \neq B$, so $A \subset B$ (strict). ✓

Step 2 - Put B as an element of C. We need $B = \{1, 2\}$ to be an *entry* of C. Take

$$C = \{\,\{1,2\},\, 3\,\}.$$

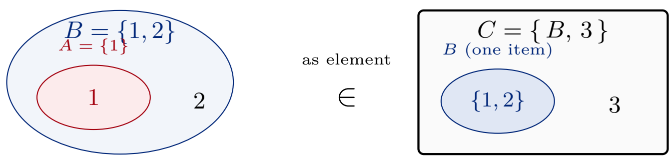
Its two entries are the set $\{1, 2\}$ and the number 3. Since $\{1, 2\} = B$ is one of them, $B \in C$. ✓(Note this is membership, not inclusion: B is an item inside C.)

Step 3 - Climb with strict inclusions. Add one fresh element at each step:

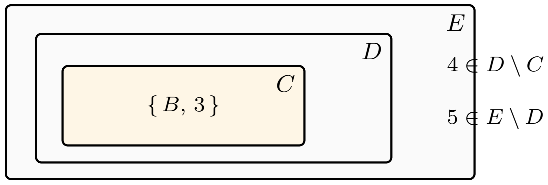
$$D = \{\,\{1,2\},\, 3,\, 4\,\}, \qquad E = \{\,\{1,2\},\, 3,\, 4,\, 5\,\}.$$

Then $C \subset D$ (all of C is in D, and $4 \in D \setminus C$) and $D \subset E$ (and $5 \in E \setminus D$). Both strict. ✓

The chain, with $\in$ and $\subset$ drawn differently. Nesting (one oval/box inside another) means $\subseteq$; a set sitting as one *loose item* in a list means $\in$. The single $\in$ link is B appearing as one element of C:



$A \subset B$ (nesting) $B \in C$ (membership)



$C \subset D \subset E$ (nesting, each adds one element)

Answer: $A = \{1\},\ B = \{1, 2\},\ C = \{\{1, 2\}, 3\},\ D = \{\{1, 2\}, 3, 4\},\ E = \{\{1, 2\}, 3, 4, 5\}.$

★ Method to remember

Transferable technique: to force "$B \in C$", list B itself as one of the elements of C (do not merge B's elements into C - that would give $B \subseteq C$ instead). To force a *strict* inclusion $X \subset Y$, take $Y = X \cup \{\text{one new element}\}$.

Problem 16 - Describe the derived sets

Problem 16

Let the universe be $U = \mathbb{Z}$, and

$$A = \{x \in \mathbb{Z} : \exists y \in \mathbb{Z}^+,\ x = 2y\}, \qquad B = \{x \in \mathbb{Z} : \exists y \in \mathbb{Z}^+,\ x = 2y - 1\},$$
$$C = \{x \in \mathbb{Z} : x < 10\}.$$

Describe the sets $\overline{A},\ \overline{A \cup B},\ \overline{C},\ A \setminus C,\ C \setminus (A \cup B)$.

The idea

First translate each defining condition into plain words, then everything is bookkeeping on the integer line. Because y ranges over the *positive* integers $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$:

- $x = 2y$ with $y \geq 1$ gives $A = \{2, 4, 6, \dots\}$, the **positive even** integers.
- $x = 2y - 1$ with $y \geq 1$ gives $B = \{1, 3, 5, \dots\}$, the **positive odd** integers.
- $C = \{x : x < 10\} = \{\dots, 7, 8, 9\}$, all integers **below** 10.

The one observation that makes the rest fall out: every positive integer is either even or odd, so $A \cup B = \mathbb{Z}^+ = \{1, 2, 3, \dots\}$. A complement $\overline{X}$ is "U minus X" ($U = \mathbb{Z}$), and a difference is "keep, then remove". Reading these off a number line makes the answers obvious.

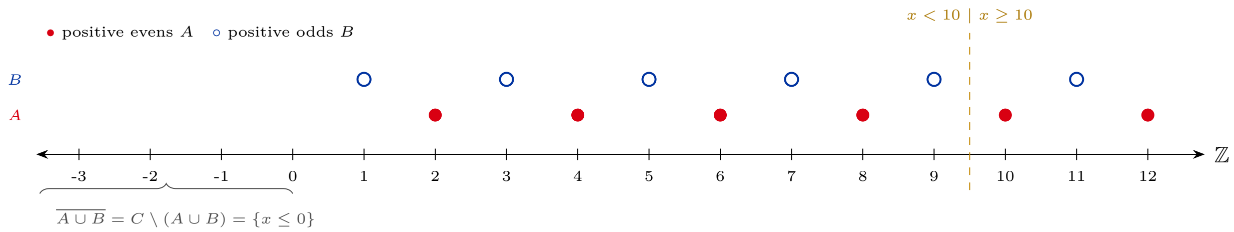
Step 1 - The two basic unions. $A \cup B$ is all positive evens together with all positive odds, i.e. *all positive integers*: $A \cup B = \mathbb{Z}^+ = \{1, 2, 3, \dots\}$.

Step 2 - Read off each set.

- $\overline{A} = \mathbb{Z} \setminus \{2, 4, 6, \dots\}$: every integer that is *not* a positive even, namely all integers $x \leq 0$ together with all positive odd numbers ($\{\dots, -2, -1, 0\} \cup \{1, 3, 5, \dots\}$).
- $\overline{A \cup B} = \mathbb{Z} \setminus \mathbb{Z}^+ = \{0, -1, -2, \dots\} = \{x \in \mathbb{Z} : x \leq 0\}$ (the non-positive integers: zero and the negatives).
- $\overline{C} = \mathbb{Z} \setminus \{x < 10\} = \{x \in \mathbb{Z} : x \geq 10\} = \{10, 11, 12, \dots\}$.
- $A \setminus C$: positive evens that are *not* below 10, i.e. positive evens ≥ 10: $\{10, 12, 14, \dots\}$.

- $C \setminus (A \cup B)$: integers below 10 with all positive integers removed, leaving the non-positive ones: $\{0, -1, -2, \dots\} = \{x \in \mathbb{Z} : x \le 0\}$. This is the *same* set as $\overline{A \cup B}$ (every $x \le 0$ is below 10, so removing $\mathbb{Z}^+$ from $\mathbb{Z}$ or from $\{x < 10\}$ leaves the same thing).

Always check (number line, window -3 to 12). The illustration below tracks A (positive evens) and B (positive odds). One can verify by eye: $A \setminus C$ starts at 10; $\overline{C}$ is everything from 10 up; and both $\overline{A \cup B}$ and $C \setminus (A \cup B)$ are exactly the points ≤ 0. A brute-force enumeration over $-20 \le x \le 20$ confirms all five descriptions and the equality $C \setminus (A \cup B) = \overline{A \cup B}$. ✓



Answer: $\overline{A} = \{x \le 0\} \cup \{1, 3, 5, \dots\}$; $\overline{A \cup B} = \{x \le 0\} = \{0, -1, -2, \dots\}$;
$\overline{C} = \{10, 11, 12, \dots\}$; $A \setminus C = \{10, 12, 14, \dots\}$; $C \setminus (A \cup B) = \{x \le 0\}$ $(= \overline{A \cup B})$.

★ Method to remember

Transferable technique: decode each set-builder into one plain sentence first ("positive evens", "integers below 10"), spot the structural fact that simplifies everything ($A \cup B = \mathbb{Z}^+$), then read complements as "U minus the set" and differences as "keep then remove". A number line turns it into something you can just look at.

Problem 20 - Three families of equivalent conditions

Problem 20

Let A, B be arbitrary subsets of a universe U. Prove that in each row, all the listed conditions are equivalent.

Row 1: $A \subseteq B \iff \overline{B} \subseteq \overline{A} \iff A \cup B = B \iff A \cap B = A$.

Row 2: $A \cap B = \varnothing \iff A \subseteq \overline{B} \iff B \subseteq \overline{A}$.

Row 3: $A \cup B = U \iff \overline{A} \subseteq B \iff \overline{B} \subseteq A$.

The idea

"These four conditions are equivalent" means: any one of them implies every other. Proving all $4 \cdot 3 = 12$ implications separately is wasteful. The efficient trick is a **cycle of implications**: prove

$$P_1 \Rightarrow P_2 \Rightarrow P_3 \Rightarrow P_4 \Rightarrow P_1.$$

Once the cycle closes, you can get from any condition to any other by walking around the loop, so all four are equivalent. We show Row 1 in full this way. For Rows 2 and 3 the three conditions in a row are, on the level of a single element x, three rephrasings of one implication (for Row 2 it is "$x \in A \Rightarrow x \notin B$"; for Row 3 it is "$x \notin A \Rightarrow x \in B$"), so a short element argument settles each row at once.

Row 1 - the cycle $A \subseteq B \Rightarrow A \cup B = B \Rightarrow A \cap B = A \Rightarrow \overline{B} \subseteq \overline{A} \Rightarrow A \subseteq B$.

(a) $A \subseteq B \Rightarrow A \cup B = B$.
Always $B \subseteq A \cup B$. Conversely, if $x \in A \cup B$ then $x \in A$ or $x \in B$; in the first case $A \subseteq B$ gives $x \in B$, in the second $x \in B$ already. So $A \cup B \subseteq B$. Both inclusions give $A \cup B = B$.

(b) $A \cup B = B \Rightarrow A \cap B = A$.
From $A \cup B = B$ and $A \subseteq A \cup B$ we get $A \subseteq B$. Then for any $x \in A$ we also have $x \in B$, so $x \in A \cap B$; hence $A \subseteq A \cap B$. The reverse $A \cap B \subseteq A$ is always true. So $A \cap B = A$.

(c) $A \cap B = A \Rightarrow \overline{B} \subseteq \overline{A}$.
$A \cap B = A$ says $A \subseteq B$ (every element of A is in $A \cap B$, hence in B). Now take $x \in \overline{B}$, i.e. $x \notin B$. If x were in A, then $x \in B$ by $A \subseteq B$ - a contradiction. So $x \notin A$, i.e. $x \in \overline{A}$. Thus $\overline{B} \subseteq \overline{A}$.

(d) $\overline{B} \subseteq \overline{A} \Rightarrow A \subseteq B$.
This is the contrapositive of (c). Take $x \in A$. Then $x \notin \overline{A}$; since $\overline{B} \subseteq \overline{A}$, also $x \notin \overline{B}$, i.e. $x \in B$. Thus $A \subseteq B$.

The cycle is closed, so all four conditions of Row 1 are equivalent. ■

Row 2 - $A \cap B = \varnothing \Leftrightarrow A \subseteq \overline{B} \Leftrightarrow B \subseteq \overline{A}$.
Read each condition for a single element x:

$$A \cap B = \varnothing \iff \text{no } x \text{ lies in both} \iff (x \in A \Rightarrow x \notin B) \iff (x \in A \Rightarrow x \in \overline{B}) \iff A \subseteq \overline{B}.$$

The middle implication "$x \in A \Rightarrow x \notin B$" is symmetric in A, B (it equals "$x \in B \Rightarrow x \notin A$", i.e. $x \in B \Rightarrow x \in \overline{A}$, i.e. $B \subseteq \overline{A}$). So all three are equivalent. ■

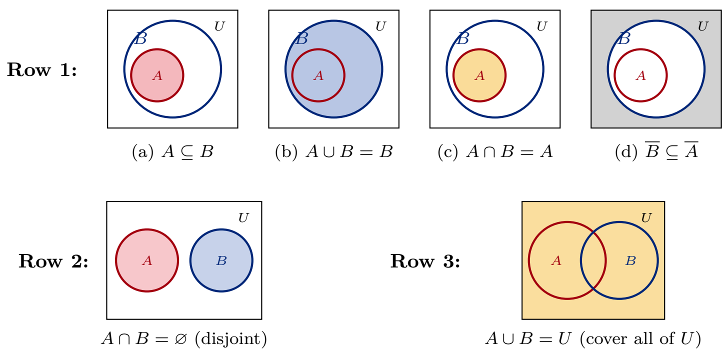
Row 3 - $A \cup B = U \Leftrightarrow \overline{A} \subseteq B \Leftrightarrow \overline{B} \subseteq A$.
Again at the element level:

$$A \cup B = U \iff \text{every } x \text{ is in } A \text{ or } B \iff (x \notin A \Rightarrow x \in B) \iff \overline{A} \subseteq B.$$

The implication "$x \notin A \Rightarrow x \in B$" is the contrapositive of "$x \notin B \Rightarrow x \in A$", which is $\overline{B} \subseteq A$. So all three are equivalent. ■

Always check. An exhaustive computer check over all pairs of subsets A, B of a 5-element universe confirms that within each row the conditions are simultaneously true or simultaneously false (no pair separates them). ✓

Seeing the equivalences. Each condition is a picture. When the four Row 1 panels all describe the *same* arrangement (A nested inside B), the equivalence is visible at a glance; likewise the single panel for each of Rows 2 and 3. (Shaded = the set named under the panel; the outer rectangle is the universe U.)



Answer: All conditions in each row are equivalent; Row 1 is proved by the implication cycle $A \subseteq B \Rightarrow A \cup B = B \Rightarrow A \cap B = A \Rightarrow \overline{B} \subseteq \overline{A} \Rightarrow A \subseteq B$, and Rows 2–3 by a single element-level implication read three ways.

★ Method to remember

Transferable technique: to show several conditions are equivalent, do not prove every pairwise implication - prove a single *cycle* $P_1 \Rightarrow P_2 \Rightarrow \cdots \Rightarrow P_n \Rightarrow P_1$. Equivalences involving complements are often cleanest via the *contrapositive* ($\overline{B} \subseteq \overline{A}$ is just $A \subseteq B$ read backwards).

Problem 54 - Two set identities (parts 3 and 11)

Problem 54

Prove the identities.

(3) $(A \cup B) \setminus C = (A \setminus C) \cup (B \setminus C)$.

(11) $A \cap (B \triangle C) = (A \cap B) \triangle (A \cap C)$, where $X \triangle Y = (X \setminus Y) \cup (Y \setminus X)$ (symmetric difference, written $\div$ in the source).

The idea

The one reliable engine for set identities is **element-chasing**: to prove $X = Y$, take a fixed x and show $x \in X \Leftrightarrow x \in Y$ by unfolding the definitions ($\cup \rightarrow$"or", $\cap \rightarrow$"and", $\setminus \rightarrow$"and not") and rearranging by simple logic. Part (3) is a clean distributive law and falls straight out. Part (11) mixes in symmetric difference; the key realization is that $x \in X \triangle Y$ means "x is in *exactly one* of X, Y" - a parity statement. After conditioning on whether $x \in A$, the "exactly one" bookkeeping lines up on both sides.

Part (3) - distributing $\setminus C$ over $\cup$. Element-chase:

$$\begin{aligned} x \in (A \cup B) \setminus C &\iff (x \in A \text{ or } x \in B) \text{ and } x \notin C \\ &\iff (x \in A \text{ and } x \notin C) \text{ or } (x \in B \text{ and } x \notin C) \quad [\text{"and" distributes over "or"}] \\ &\iff x \in (A \setminus C) \text{ or } x \in (B \setminus C) \\ &\iff x \in (A \setminus C) \cup (B \setminus C). \end{aligned}$$

Since this holds for every x, the two sets are equal. ■

Part (11) - intersection distributes over symmetric difference. Recall $x \in X \triangle Y \iff x$ belongs to *exactly one* of X, Y. Fix an arbitrary x and split on membership in A.

Case $x \notin A$. Then $x \notin A \cap (B \triangle C)$ (left side). Also $x \notin A \cap B$ and $x \notin A \cap C$, so x is in *neither* $A \cap B$ nor $A \cap C$, hence $x \notin (A \cap B) \triangle (A \cap C)$ (right side). Both sides exclude x. ✓

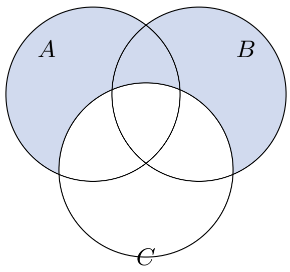
Case $x \in A$. Then $x \in A \cap B \iff x \in B$ and $x \in A \cap C \iff x \in C$. Therefore

$$\begin{aligned} x \in (A \cap B) \triangle (A \cap C) &\iff x \text{ is in exactly one of } A \cap B,\ A \cap C \\ &\iff x \text{ is in exactly one of } B,\ C \quad [x \in A, \text{ so } A \cap \text{ is automatic}] \\ &\iff x \in B \triangle C \\ &\iff x \in A \cap (B \triangle C) \quad [\text{because } x \in A]. \end{aligned}$$

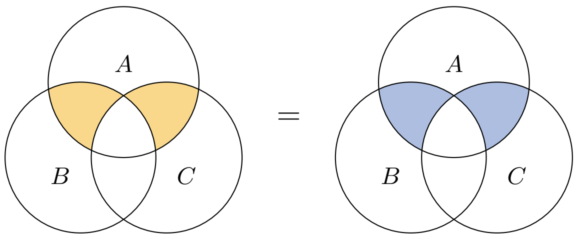
So on $\{x \in A\}$ the two sides agree as well. As both cases match, the sets are equal. ■

Always check. Exhaustive enumeration over all triples A, B, C of subsets of a 4-element ground set finds *zero* disagreements for both identities. A Venn check for (3): the region "in A or B, outside C" is precisely the two crescents "in A, outside C" and "in B, outside C" taken together. ✓

$(A \cup B) \setminus C$ is shaded



Part (11) as a picture. Both sides shade the *same* region: the two slivers of A that lie in exactly one of B, C. Three circles A (top), B (lower left), C (lower right):



Answer: Both identities hold. (3): "in A or B, not in C" = "in A not C" or "in B not C". (11): after fixing whether $x \in A$, "exactly one of B, C" matches "exactly one of $A \cap B, A \cap C$".

★ Method to remember

Transferable technique: prove $X = Y$ by element-chasing - unfold $\cup, \cap, \setminus$ into "or / and / and-not" and simplify the logic. For anything with $\triangle$, replace it by "in exactly one of the two" (a parity statement) and, if a variable like A appears on both sides, split into the cases $x \in A$ and $x \notin A$.

Problem 58 - $A \times B = B \times A$ forces $A = B$

Problem 58

Prove that $A \times B = B \times A$ implies $A = B$.

The idea

An element of a product is an *ordered* pair, and order is what carries the information. If $A \times B = B \times A$, then any pair you can form one way must already be available the other way, and matching the first coordinate of $(a, b) \in A \times B$ against the first coordinate of pairs in $B \times A$ forces $a \in B$. To prove the set equality $A = B$ we show two inclusions $A \subseteq B$ and $B \subseteq A$.

One honest caveat the statement hides: it needs A and B *nonempty*. If either is empty, both products are empty and the hypothesis holds for free, telling us nothing - so the conclusion can fail. We flag the counterexample, then prove the claim under the (intended) nonempty assumption.

Step 0 - The empty-set edge case. Take $A = \varnothing$ and $B = \{1\}$. Then $A \times B = \varnothing$ and $B \times A = \varnothing$, so $A \times B = B \times A$ holds, yet $A \neq B$. So the implication can fail only when *exactly one* of A, B is empty (if both are empty it holds trivially, since then $A = B = \varnothing$); assume $A \neq \varnothing$ and $B \neq \varnothing$ from now on, the only case left to argue.

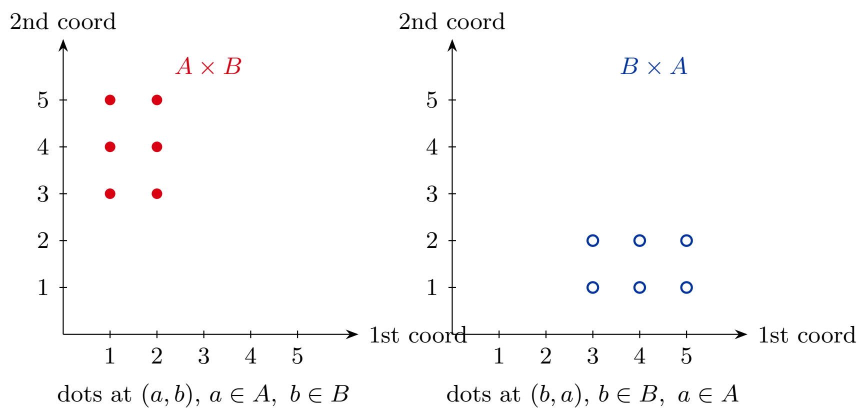
Step 1 - Show $A \subseteq B$. Let $a \in A$ be arbitrary. Since $B \neq \varnothing$, pick some $b \in B$. Then $(a, b) \in A \times B$. By hypothesis $A \times B = B \times A$, so $(a, b) \in B \times A$. An element of $B \times A$ has its *first* coordinate in B; that first coordinate is a, so $a \in B$. As $a \in A$ was arbitrary, $A \subseteq B$.

Step 2 - Show $B \subseteq A$. Symmetrically, let $b \in B$ be arbitrary. Since $A \neq \varnothing$, pick some $a \in A$. Then $(a, b) \in A \times B = B \times A$, and an element of $B \times A$ has its *second* coordinate in A; that second coordinate is b, so $b \in A$. Hence $B \subseteq A$.

Step 3 - Conclude. $A \subseteq B$ and $B \subseteq A$ give $A = B$. ■

Always check. Over all nonempty subsets A, B of $\{0, 1, 2\}$, an exhaustive search finds no pair with $A \times B = B \times A$ but $A \neq B$; and the empty case $A = \varnothing$, $B = \{1\}$ does satisfy the hypothesis with $A \neq B$, exactly as flagged. ✓

Why the swap forces $A = B$. Plot each pair as a dot. With $A = \{1, 2\}$, $B = \{3, 4, 5\}$ the two products $A \times B$ (dots at (a, b)) and $B \times A$ (dots at (b, a)) land on *different* points - they sit in different parts of the plane and cannot coincide unless A and B are the same set of numbers:



The red and blue point sets are reflections across the diagonal; they overlap only on the diagonal, so $A \times B = B \times A$ can hold only when $A = B$.

Answer: For nonempty A, B: $A \times B = B \times A \Rightarrow A = B$, proved by the two inclusions. The nonemptiness is needed: $A = \varnothing$, $B = \{1\}$ is a counterexample without it.

★ Method to remember

Transferable technique: to prove a set equality $A = B$, prove $A \subseteq B$ and $B \subseteq A$ separately ("take an arbitrary element, show it lies in the other"). When products appear, read off the coordinate that carries the information: the first coordinate of a pair in $X \times Y$ lives in X, the second in Y. And always test an "obvious" implication on the empty set before trusting it.

Problem 60 - Cartesian-product distributive laws

Problem 60

Prove the identities (1)–(6):

(1) $A \times (B \cup C) = (A \times B) \cup (A \times C)$;

(2) $(B \cup C) \times A = (B \times A) \cup (C \times A)$;

(3) $A \times (B \cap C) = (A \times B) \cap (A \times C)$;

(4) $(B \cap C) \times A = (B \times A) \cap (C \times A)$;

(5) $A \times (B \setminus C) = (A \times B) \setminus (A \times C)$;

(6) $(B \setminus C) \times A = (B \times A) \setminus (C \times A)$.

The idea

Every element of a product is an ordered pair (x, y), and "$(x, y) \in X \times Y$" is just the conjunction "$x \in X$ *and* $y \in Y$". So the moment you write (x, y) and unfold both sides into statements about x and y separately, each identity becomes a tiny piece of propositional logic: $\cup$ is "or" on the y-part, $\cap$ is "and", $\setminus$ is "and not", and the fixed factor $x \in A$ just rides along unchanged. Prove (1), (3), (5) by this ordered-pair iff method; (2), (4), (6) are the *same* proofs with the product on the other side - swap the roles of the coordinates.

(1) Union on the right factor.

$$\begin{aligned}
(x,y) \in A \times (B \cup C) &\iff x \in A \text{ and } (y \in B \text{ or } y \in C) \\
&\iff (x \in A \text{ and } y \in B) \text{ or } (x \in A \text{ and } y \in C) \quad \text{[distribute "and" over "or"]} \\
&\iff (x,y) \in A \times B \text{ or } (x,y) \in A \times C \\
&\iff (x,y) \in (A \times B) \cup (A \times C). \qquad \blacksquare
\end{aligned}$$

(3) Intersection on the right factor. Same chase with "and":

$$\begin{aligned}
(x,y) \in A \times (B \cap C) &\iff x \in A \text{ and } y \in B \text{ and } y \in C \\
&\iff (x \in A \text{ and } y \in B) \text{ and } (x \in A \text{ and } y \in C) \quad [x \in A \text{ duplicated freely}] \\
&\iff (x,y) \in (A \times B) \cap (A \times C). \qquad \blacksquare
\end{aligned}$$

(5) Difference on the right factor.

$$(x,y) \in A \times (B \setminus C) \iff x \in A \text{ and } y \in B \text{ and } y \notin C.$$

On the other side,

$$(x,y) \in (A \times B) \setminus (A \times C) \iff (x \in A \text{ and } y \in B) \text{ and } \neg(x \in A \text{ and } y \in C).$$

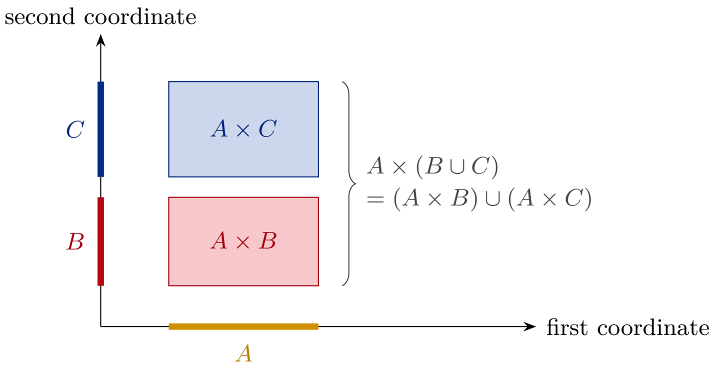
But the first clause already gives $x \in A$, so $\neg(x \in A \text{ and } y \in C)$ collapses to $y \notin C$. Hence this is $x \in A$ and $y \in B$ and $y \notin C$ - identical to the line above. So the two sets are equal. $\blacksquare$

(2), (4), (6) - the mirror images. These are (1), (3), (5) with the variable factor A moved to the *first* coordinate. Run the identical argument on a pair (y, x) with $(y, x) \in X \times A \iff y \in X$ and $x \in A$; the fixed part is now $x \in A$ in the second slot, and $\cup, \cap, \setminus$ distribute over the y-part exactly as before. Concretely:

$$\begin{aligned}
(y,x) \in (B \cup C) \times A &\iff (y \in B \text{ or } y \in C) \text{ and } x \in A \iff (y,x) \in (B \times A) \cup (C \times A), \\
(y,x) \in (B \cap C) \times A &\iff y \in B \text{ and } y \in C \text{ and } x \in A \iff (y,x) \in (B \times A) \cap (C \times A), \\
(y,x) \in (B \setminus C) \times A &\iff y \in B \text{ and } y \notin C \text{ and } x \in A \iff (y,x) \in (B \times A) \setminus (C \times A). \qquad \blacksquare
\end{aligned}$$

Always check. All six identities were verified by exhaustive enumeration over every triple A, B, C of subsets of $\{0, 1, 2\}$ (so every possible product was compared elementwise); *zero* mismatches. ✓

The distributive law (1) as an area picture. Draw A as an interval on the x-axis and B, C as two stacked bands on the y-axis. A product $X \times Y$ is the rectangle "above X, beside Y". Then $A \times (B \cup C)$ is the full block over A spanning B and C, which is exactly the $A \times B$ rectangle stacked on the $A \times C$ rectangle:



Answer: All six hold. The right-factor laws (1),(3),(5) follow from $(x, y) \in X \times Y \iff x \in X \wedge y \in Y$, distributing the set operation over the y-coordinate while $x \in A$ rides along; the left-factor laws (2),(4),(6) are the same with the coordinates swapped.

★ Method to remember

Transferable technique: a Cartesian product turns set algebra into logic on coordinates - $(x, y) \in X \times Y \iff (x \in X) \wedge (y \in Y)$. To prove a product identity, write a generic pair (x, y), expand both sides into and/or/not statements about the two coordinates, and use ordinary logical distribution. A constant factor (here A) simply appears, unchanged, in the same coordinate on both sides.